

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

|                  |                                                                                                                                  |               |                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------|
| Course Code:     | DSA03                                                                                                                            | Course Title: | Natural Language Processing                          |
| CO Assessed:     | CO5 – Articulation                                                                                                               | Assessment:   | Assessment Tool 1 – Constraint-Based Problem Solving |
| Weightage:       | 35% of CIA                                                                                                                       |               |                                                      |
| CO5 Statement:   | Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5) |               |                                                      |
| Student Name:    | P.Hithaishi                                                                                                                      | Reg. No.:     | 192425196                                            |
| Section / Batch: | 2024-C                                                                                                                           | Date:         | 25-8-2026                                            |

### Code of Conduct

*I P. Hithaishi (192425196) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: P. Hithaishi

### Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

| Section / Topic                                                             | Focus                                                                                                                                                                                                | Question No. |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Discourse, Reference Resolution & Text Coherence                            | Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.       | Q1           |
| Dialog and Conversational Agents, Discourse Planning & Surface Realizations | Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.           | Q2           |
| Machine Translation, Discourse Structure, Predicate Logic & Interpretation  | Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases. | Q3           |

### **1. Reference Resolution and Discourse Coherence**

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility
- Discourse coherence

c) Construct a table showing:

| Referring Expression | Possible Antecedents | Valid/Invalid | Reason |
|----------------------|----------------------|---------------|--------|
|----------------------|----------------------|---------------|--------|

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

### **2. Dialog Act Recognition and Response Generation**

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
  - review
  - improve
  - confident
  - feedback
4. Response length must be 2–3 sentences.

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

**Tasks:**

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
  - Coherence
  - Constraint satisfaction
  - Politeness
  - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

### **3. Word Sense Disambiguation and Meaning Representation**

**Source Sentence:**

**"Priya sat on the bank and watched the boats moving across the water."**

**Note:**

**The word "bank" can refer to:**

- A financial institution
- The side of a river

**Constraints:**

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

**Tasks:**

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

**sit(x)**

**bank(y)**

**beside(x,y)**

**watch(x,z)**

**boat(z)**

**move(z)**

**water(w)**

**d) Generate a simple English paraphrase without ambiguity.**

**e) Explain how Word Sense Disambiguation improves Machine Translation.**

## Rubrics for Calculation

| Criteria                | Weightage | Level 4<br>(Exemplary)                                                         | Level 3<br>(Proficient)                               | Level 2<br>(Developing)                       | Level 1<br>(Beginning)                   |
|-------------------------|-----------|--------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|------------------------------------------|
| Problem Understanding   | 25%       | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps           | Partial understanding; misses key constraints | Fails to identify constraints correctly  |
| Constraint Analysis     | 25%       | Analyzes constraints effectively and prioritizes them logically                | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization         | No meaningful analysis or prioritization |
| Solution Development    | 25%       | Develops optimal and feasible solutions satisfying all constraints             | Develops workable solutions with minor limitations    | Solutions partially satisfy constraints       | Solutions do not meet constraints        |
| Application of Concepts | 15%       | Effectively applies appropriate concepts, methods, or tools                    | Applies concepts with minor errors                    | Limited application; requires guidance        | Unable to apply relevant concepts        |
| Justification           | 10%       | Provides strong justification for decisions with logical reasoning             | Provides justification with some clarity              | Weak or unclear justification                 | No justification provided                |

| Q. No. / Criteria Level | Problem Understanding | Constraint Analysis | Solution Development | Application of Concepts | Justification | Sub. Total | Total % |
|-------------------------|-----------------------|---------------------|----------------------|-------------------------|---------------|------------|---------|
| 1                       | 7                     | 7                   | 7                    | 6                       | 6             | 33         | 94      |
| 2                       | 7                     | 7                   | 7                    | 6                       | 6             | 33         |         |
| 3                       | 6                     | 6                   | 6                    | 5                       | 5             | 28         |         |

| Faculty In-charge      | Course Coordinator | Program Director |
|------------------------|--------------------|------------------|
| Dr. T. Kumaragurubaran |                    |                  |
| Signature: _____       | Signature: _____   | Signature: _____ |
| Date: _____            | Date: _____        | Date: _____      |

## 1. Reference Resolution and Discourse Coherence

### Given paragraph:

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

#### a) Referring expressions and possible antecedents

| Referring Expression | Possible Antecedents            |
|----------------------|---------------------------------|
| He                   | Ravi, Arun                      |
| him                  | Ravi, Arun                      |
| it                   | Book on Artificial Intelligence |
| they                 | Ravi and Arun                   |
| the book             | Book on Artificial Intelligence |

#### b & c) Applying constraints

| Referring Expression | Possible Antecedents | Valid/Invalid | Reason                                                                                               |
|----------------------|----------------------|---------------|------------------------------------------------------------------------------------------------------|
| He                   | Ravi                 | Valid         | Ravi is a singular male; grammatical agreement is satisfied.                                         |
| He                   | Arun                 | Valid         | Arun is also a singular male, so gender/number agreement is satisfied.                               |
| him                  | Ravi                 | Valid         | Ravi is male and singular; semantically, Ravi can be helped.                                         |
| him                  | Arun                 | Invalid       | Arun is the person performing the helping action, so referring to himself as “him” is less coherent. |
| it                   | Book                 | Valid         | “It” is singular and refers naturally to the book.                                                   |
| it                   | Ravi/Arun            | Invalid       | People are normally referred to using “him/he,” not “it.”                                            |
| they                 | Ravi + Arun          | Valid         | “They” is plural and refers naturally to both people.                                                |
| the book             | AI book              | Valid         | Recency and semantic context indicate the previously mentioned book.                                 |

#### Important point about “He”

“He” is ambiguous by gender and number alone, because both Ravi and Arun are male and singular.

However, recency favors Arun, because Arun is the most recently mentioned person before “He.”

So:

**He** → **Arun** is the most likely resolution.

#### **d) Final coreference chains**

The most coherent interpretation is:

- Arun → He → him
- Ravi → they
- Arun → they
- Book on Artificial Intelligence → it → the book

So the chains are:

**Chain 1:** Arun → He → him

**Chain 2:** Ravi + Arun → they

**Chain 3:** Book on Artificial Intelligence → it → the book

Note: “He” is technically ambiguous, but under the given constraints, Arun is the preferred antecedent because of recency and discourse coherence.

#### **e) Rewrite the paragraph with references**

##### **Rewritten paragraph:**

Ravi met Arun at the library. Arun was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

This version removes all pronoun ambiguity.

#### **f) Ambiguity if “He was looking for a book” appears before Arun is introduced**

If the sentence were:

**“Ravi met at the library. He was looking for a book. Arun arrived later.”**

or if **“He was looking for a book”** appeared before Arun was introduced, then **He** could only refer to Ravi if Ravi had already been introduced.

If neither Ravi nor Arun had been introduced, **“He” would have no clear antecedent.**

Therefore, reference resolution requires a previously introduced entity and sufficient contextual information.

##### **Short exam answer:**

If “He was looking for a book” occurs before Arun is introduced, “He” cannot refer to Arun because Arun is not yet available as an antecedent. This may cause ambiguity or an unresolved reference. The system would rely on discourse context to determine the intended person.

## 2. Dialog Act Recognition and Response Generation

### Given conversation:

**Student:** “I submitted my assignment, but I am worried that I may have made many mistakes.”

#### a) Required dialog acts

The required dialog acts are:

1. **Reassure** – reduce the student's worry.
2. **Advise** – provide useful guidance about the assignment.

So:

Dialog Act = Reassure + Advise

#### b) important entities

The important entities that should remain coherent are:

- **assignment**
- **mistakes**
- **you**

The response should continue talking about the student's assignment and possible mistakes, rather than suddenly changing the topic.

#### c) Three possible responses

##### Response 1

Don't worry, you have already completed the assignment, and mistakes are a normal part of learning. You can review the assignment and use feedback to improve your work and feel more confident next time.

##### Response 2

It is completely normal to worry about mistakes after submitting an assignment, so you should not be too hard on yourself. When you receive feedback, review your mistakes carefully so you can improve and become more confident.

##### Response 3

You have done your part by submitting the assignment, so try to stay confident. If there are mistakes, use the feedback to review them and improve your future assignments.

#### d) Comparison of responses

| Criteria                | Response 1          | Response 2              | Response 3              |
|-------------------------|---------------------|-------------------------|-------------------------|
| Coherence               | Excellent           | Excellent               | Excellent               |
| Constraint satisfaction | Excellent           | Excellent               | Excellent               |
| Politeness              | Positive and polite | Positive and supportive | Positive and supportive |
| Relevance               | Highly relevant     | Highly relevant         | Highly relevant         |



|        |                   |                   |                   |
|--------|-------------------|-------------------|-------------------|
| Advice | Review + feedback | Review + feedback | Review + feedback |
|--------|-------------------|-------------------|-------------------|

#### e) Best response

##### Best choice: Response 2

It is completely normal to worry about mistakes after submitting an assignment, so you should not be too hard on yourself. When you receive feedback, review your mistakes carefully so you can improve and become more confident.

##### Justification:

Response 2 is the best because:

- It reassures the student.
- It provides clear advice.
- It maintains entity coherence using assignment/mistakes/you.
- It contains the required keywords feedback, review, improve, confident.
- It has a clear cause–effect relation: feedback → review mistakes → improve → become confident.
- It is positive, polite, and relevant.
- It does not give contradictory advice.

#### f) What happens if entity coherence is not maintained?

If entity coherence is not maintained, the response may become confusing or irrelevant.

For example:

“Don't worry about your assignment. You should review your car and use feedback to improve your **project presentation.**”

Here, the topic suddenly changes from the assignment and mistakes to a car and presentation. This breaks discourse coherence and makes the response difficult to understand.

##### Exam answer:

When entity coherence is not maintained, the system may lose track of the main topic or entities. This can produce irrelevant, confusing, or logically inconsistent responses.

### 3. Word Sense Disambiguation and Meaning Representation

#### Source sentence:

“Priya sat on the bank and watched the boats moving across the water.”

#### a) Possible meanings of “bank”

The word **bank** has two possible meanings:

1. **Financial institution** – a place where people deposit or withdraw money.
2. **Side of a river** – the land beside a river or body of water.

## b) Correct sense and justification

### Correct sense:

Bank = side of a river

### Why?

The surrounding words provide strong contextual clues:

- **sat on the bank**
- **boats**
- **water**
- **moving across the water**

These words are strongly associated with a riverbank.

A financial bank does not normally have boats moving across water beside it.

Therefore:

bank = riverbank / side of a river

This is an example of Word Sense Disambiguation (WSD), where the correct meaning of a word is selected using its surrounding context.

## c) Predicate logic representation

We can represent the sentence as:

sit (Priya, bank)

bank(bank)

beside (Priya, bank)

watch (Priya, boats)

boat(boats)

move(boats)

water(water)

A more complete representation is:

$$\begin{aligned} & \exists x \exists y \exists z \exists w \\ & \text{Priya}(x) \wedge \text{bank}(y) \wedge \text{sit}(x, y) \wedge \text{beside}(x, y) \\ & \wedge \text{boat}(z) \wedge \text{water}(w) \wedge \text{watch}(x, z) \\ & \wedge \text{move}(z) \wedge \text{across}(z, w) \end{aligned}$$

### Meaning:

- **Priya(x)** → x is Priya
- **bank(y)** → y is a riverbank
- **sit(x,y)** → Priya sits on the bank
- **boat(z)** → z is a boat
- **watch(x,z)** → Priya watches the boats
- **move(z)** → boats are moving
- **water(w)** → w is water
- **across(z,w)** → boats move across the water

### d) Simple English paraphrase

**Priya sat beside the river and watched the boats moving across the water.**

This removes the ambiguity of the word bank.

### e) How WSD improves Machine Translation

Word Sense Disambiguation helps Machine Translation by selecting the correct meaning of an ambiguous word before translating it.

For example:

“Priya sat on the bank.”

If bank is incorrectly interpreted as a financial institution, the translation may produce a wrong meaning.

After WSD:

**bank → riverbank**

the translation system can produce a sentence meaning:

“Priya sat beside the river.”